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Who may read our messages?

Privacy vs. state surveillance of communication – with Chat Control as a case study

[Group members]

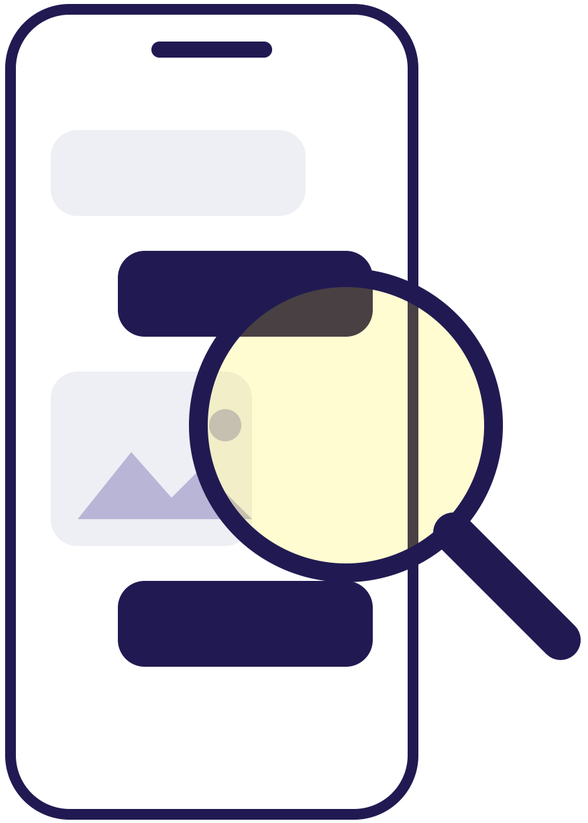
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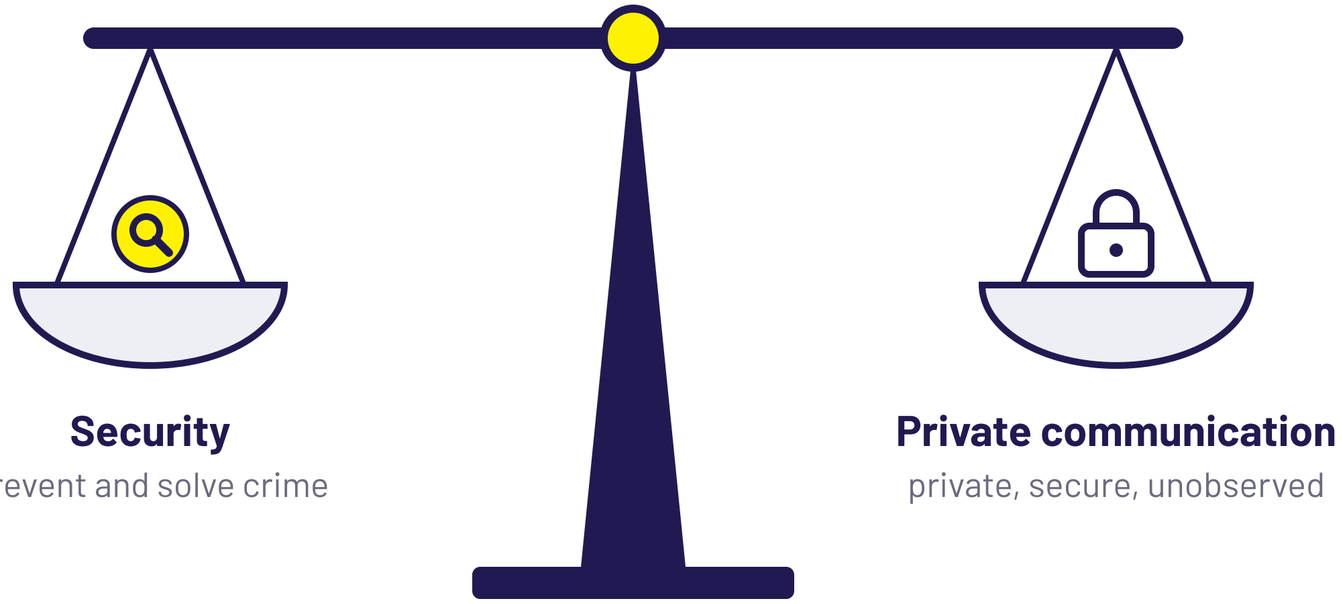
Who should be able to read your messages?

Most people would say: me, the person I write to, and in some cases the police with a court order.

Where that line is drawn has been argued about for more than 30 years.



Two legitimate goals on one scale



Security
prevent and solve crime

Private communication
private, secure, unobserved

The state has a duty to protect people, for example children, from serious crime.

People have a right to private communication (EU Charter, Art. 7 and 8).

More access for the state means less confidentiality, and the other way round. Every option gives something up.

The same conflict comes back with every new technology

1993
Clipper Chip: US plan for keys held by the state

2006
EU data retention: store everyone's call data

2013
Snowden reveals NSA mass surveillance

2014
EU court strikes down data retention

2016
Apple refuses to unlock an iPhone for the FBI

2020
Police hack the EncroChat network

2022
EU Commission proposes Chat Control 2.0

2025
UK order: Apple drops encrypted backups

Four ways the state can listen

	Targeted: one suspect	Mass: everyone
Content	**Wiretap, state trojan, spyware** What a suspect says or writes. Germany 2024: 16,868 wiretap orders. Needs a judge's order.	**Bulk interception, scanning** Everyone's content is searched. NSA programmes (Snowden, 2013); **Chat Control 2.0** (proposed 2022).
Metadata	**Call records of a suspect** Who, when, where — for one person. e.g. "quick freeze" of data after a crime.	**Data retention** Everyone's connection data is stored in advance. EU law struck down 2014; Germany debates IP storage (2026).

Encryption moved the point of access to our phones

Before



Provider can read the message; police ask the provider.

End-to-end



Provider only sees scrambled data. Investigators call this “going dark”.

The state’s three options today

1 · Hack the device
state trojans, spyware

2 · Scan on the device
client-side scanning (Chat Control)

3 · Weaken encryption
backdoors (UK vs. Apple, 2025)

Part 2

Why privacy matters

Arguments against more surveillance

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“I’ve got nothing to hide”

The most common argument for surveillance, and three problems with it (Solove, 2007)



Small facts add up

Harmless data points, combined, show health, beliefs, relationships and routines.



It is about power

Whoever holds the data decides how it is used. The person concerned cannot check or correct it.

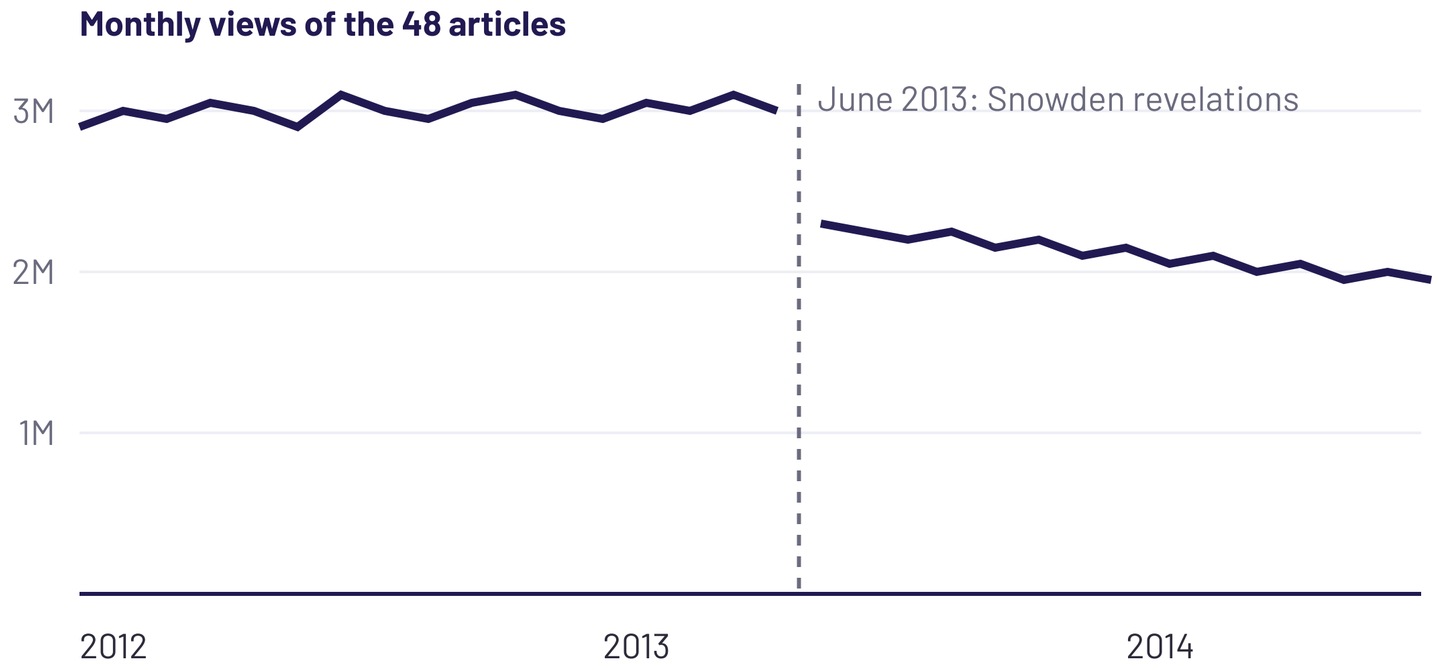


Rules change

What is harmless today can become a problem under a future government or in another country.

Solove, D. J. (2007). “I’ve Got Nothing to Hide” and Other Misunderstandings of Privacy. San Diego Law Review, 44, 745.

People change their behaviour when they feel watched



–20%

monthly views of 48 Wikipedia articles on terrorism-related topics after the Snowden revelations in June 2013.

Views kept falling in the following months. Researchers call this a **chilling effect**.

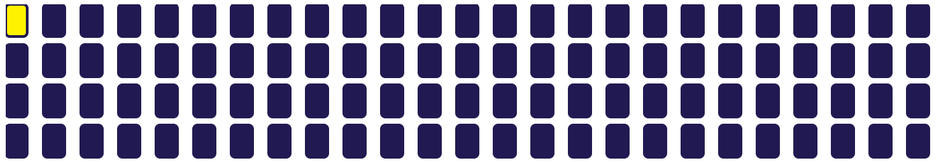
Penney, J. (2016). Chilling Effects: Online Surveillance and Wikipedia Use. Berkeley Technology Law Journal, 31, 117. Chart simplified.

Access built for the police can be used by others

Greece, 2004–05

The Athens Affair

Unknown attackers used the legal wiretap function in Vodafone Greece's network to listen to about **100 phones**, including the Prime Minister's.



USA and worldwide, 2024–25

Salt Typhoon

Chinese state hackers got into US telecom networks, including systems used for court-ordered wiretaps. In 2025, 13 countries reported **200+ US organisations** affected and **80+ countries** targeted.

In Dec 2024 the FBI and CISA advised Americans to use **encrypted messengers**.

Security researchers conclude that a way in that works only for authorities cannot be built (Abelson et al., 2015).

Prevelakis & Spinellis (2007), IEEE Spectrum; joint advisory of 13 countries (27 Aug 2025); Abelson et al. (2015), Keys under doormats

Surveillance tools spread beyond their original purpose



50,000

phone numbers on a leaked list of possible **Pegasus** spyware targets (2021)

~90

WhatsApp users warned in 2025 about **Paragon** spyware, incl. journalists in Italy

2025

German Constitutional Court bans state trojans for **minor offences**

Pegasus Project (Forbidden Stories / Amnesty, 2021); TechCrunch (2026) on Paragon; BVerfG 1 BvR 180/23 (2025)

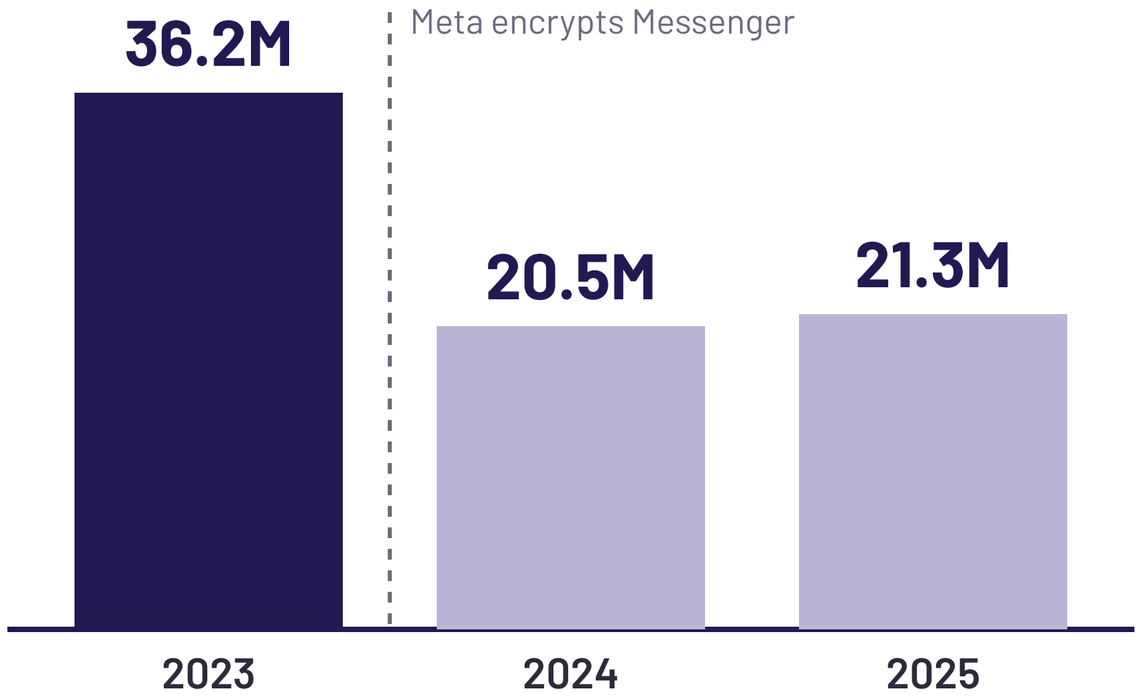
Part 3

Why the state wants access

Arguments for more surveillance

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Encryption also hides serious crime



–43%

fewer reports of child sexual abuse material from 2023 to 2024. A large part of the drop came after Facebook Messenger switched to end-to-end encryption.

Investigators argue that they cannot act on crimes they cannot see.

Reports to NCMEC, the US hotline most platforms report to (CyberTipline data 2023–2025; NBC News, 2025)

Targeted operations against encrypted networks

Police hacked or secretly ran encrypted phone services used mainly by criminals

EncroChat, 2020

6,558

arrests across Europe, about €900M seized. Around 60,000 users.

ANOM, 2021

800+

arrests in 16 countries. The FBI secretly ran the app on 12,000 devices.

Sky ECC, 2021

585

arrest warrants in Germany alone (until 2025). About 70,000 users monitored.

These cases show that investigators can reach encrypted communication without scanning everyone. Courts are still debating how such evidence may be used (CJEU, C-670/22, 2024).

Europol / Eurojust (2023); FBI & Australian Federal Police (2021); BKA (2025)

Part 4

Case study: Chat Control

The EU plan to scan private chats for child abuse material

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Chat Control 1.0 and 2.0

Since 2021

1.0 · Companies may scan

A temporary EU exception lets Meta, Google, Microsoft and others search messages voluntarily.

It expired in April 2026 and was renewed in July 2026 until 2028, now **without encrypted chats**.

Proposed 2022

2.0 · Companies must scan

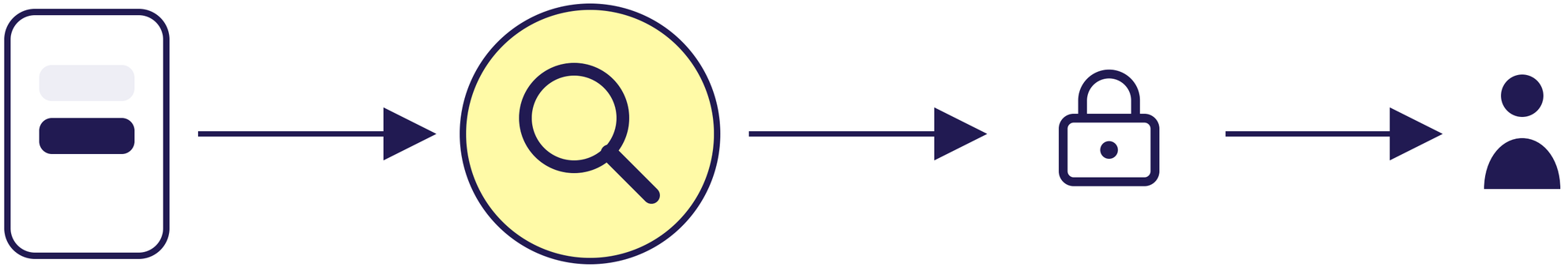
A permanent law. Authorities could order a service to scan all its users' messages, **including encrypted ones**.

A new EU Centre would collect reports and pass them on to the police.

In the matrix from page 5, version 2.0 belongs in the field "mass surveillance of content".

Regulation (EU) 2021/1232; Regulation (EU) 2026/1881; European Commission, COM(2022) 209

A scanner on every phone, before encryption



1. You write
photo, link or text

2. Your phone checks it
known images: list of fingerprints
new images and chats: AI

3. Encrypted & sent
as usual

Your friend

If flagged: report → EU Centre → police

The encryption stays technically intact, but the content has already been read on the device. Security researchers therefore treat client-side scanning like a backdoor (Abelson et al., 2024).

Four years of negotiations, no agreement yet

May 2022
Commission: scanning mandatory

Nov 2023
Parliament: only suspects, judge's order

Nov 2025
Member states: voluntary scanning

Sep 2026
6th round: no deal

Dec 2026?
next round reported

Agreed

Risk assessment for platforms, removal and blocking orders, the EU Centre.

Still open

Scanning orders for private chats, encrypted services, age verification.

EP negotiating team (29 Jun 2026); Council doc. 11956/26 (10 Sep 2026); netzpolitik.org (2 Oct 2026). Status: 27 Oct 2026.

Who stands where

← Scan everyone

Scan no one →



EU Commission

2022 proposal:
all chats, incl. encrypted

+ child-protection NGOs, Europol

Member states

voluntary scanning by companies

+ Meta, Google, Microsoft

EU Parliament

only suspects, with a judge's order

+ Germany, data-protection authorities

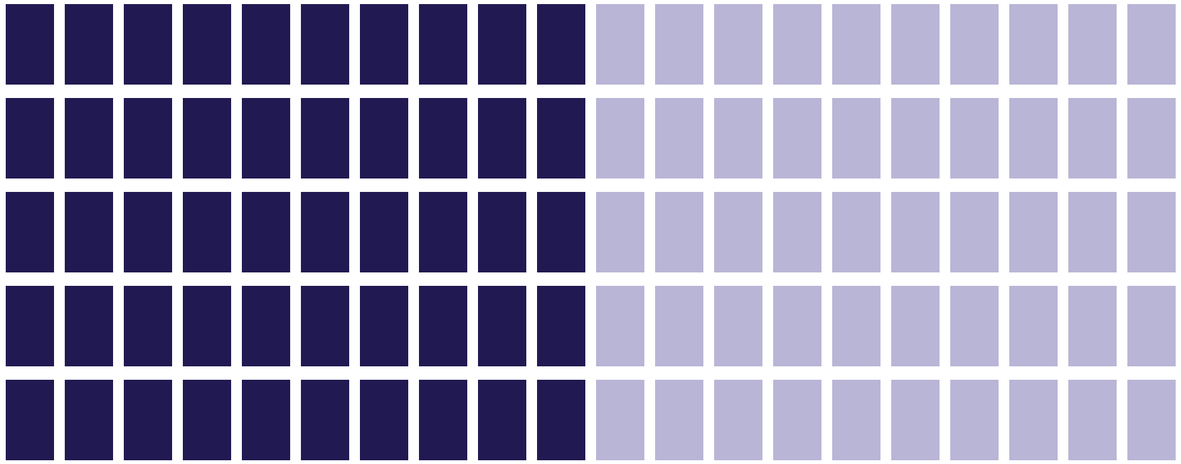
Signal & scientists

no scanning of private chats

~500 researchers; Signal would leave the EU

Many alarms concern legal content

Reports to German police (BKA), 2023



criminally relevant **not** relevant, e.g. teens' own photos, memes

~ 1 in 2

reports in Germany was not a crime

1 in 9

Irish reports (2020) were not abuse material; the police kept the data anyway

BKA figures via netzpolitik.org (2024); Irish police figures via RTÉ (2022). Each square = 1% of 180,287 reports.

Part 5

Weighing the dilemma

One framework from the course

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Five questions to weigh the dilemma

Principlist framework for cybersecurity ethics — Formosa, Wilson & Richards (2021), Lecture 2



Beneficence

Does it **do good**?



Non-maleficence

Does it **cause harm**?



Autonomy

Does it respect people's **freedom**?



Justice

Is it **fair** to everyone?

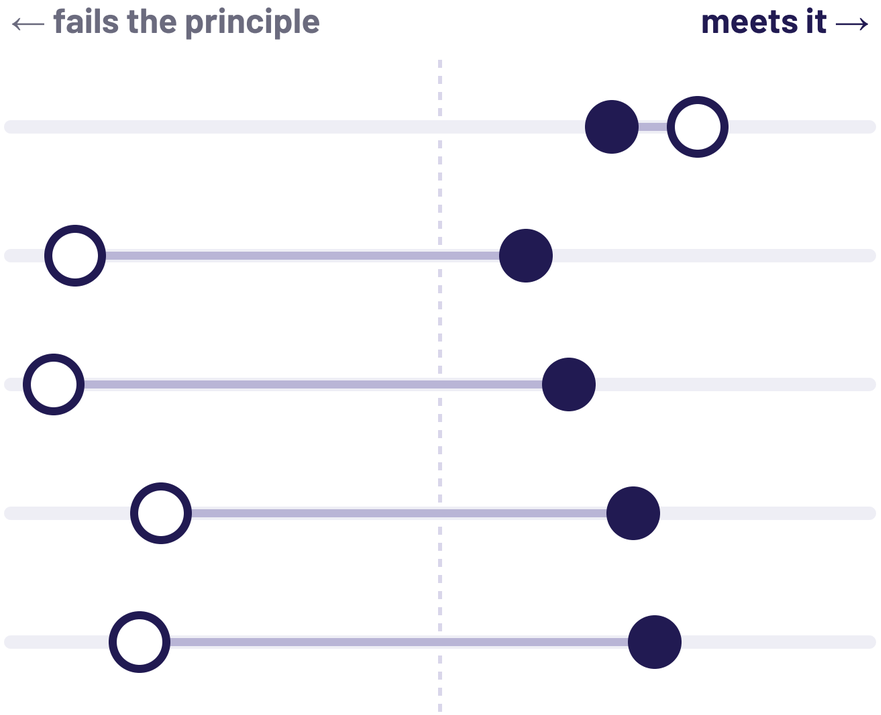


Explicability

Is it **transparent** and accountable?

We use the five questions to compare two approaches: scanning everyone (Chat Control 2.0) and targeted measures against suspects.

Mass scanning vs. targeted measures



both find offenders; mass scanning finds more

mass: weaker security, innocent people flagged

mass: everyone watched, chilling effect

mass: every user treated as a suspect

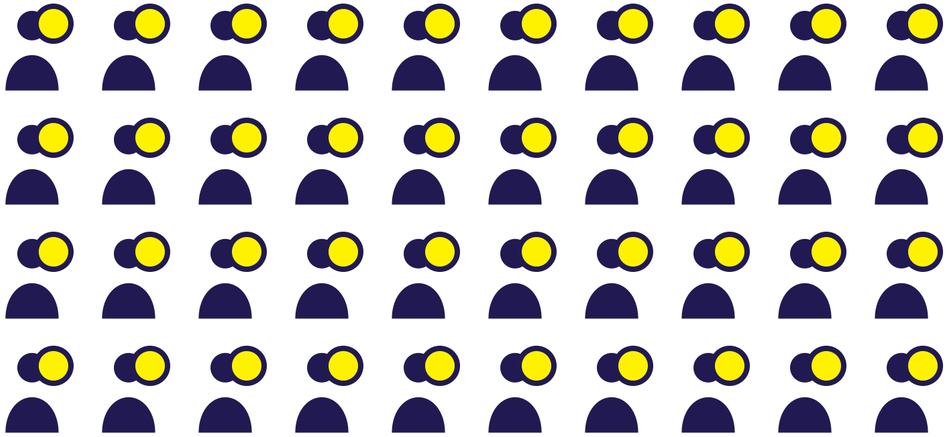
targeted: judge's order, documented

○ Mass scanning (Chat Control 2.0) ● Targeted measures (judge's order)

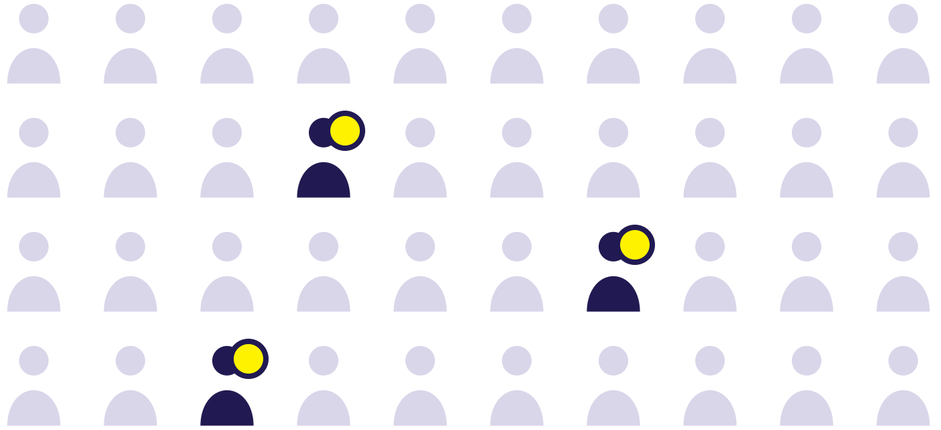
Positions show our group's assessment, based on the evidence in this presentation.

Check everyone, or only suspects?

Mass scanning
everyone is checked, all the time



Targeted
only suspects, with a judge's order



The European Parliament supports the targeted model, while most member states want wider options. The negotiations have not settled this yet.

What we conclude

1 **The conflict is old.** From the Clipper Chip to Chat Control, each new technology has raised the same question again.

2 **Encryption moved it onto our devices.** Hacking, scanning and backdoors all reach into the phone itself.

3 **Targeted measures do better.** In our assessment they meet four of the five principles better than scanning everyone, and they still catch criminals.

4 **Infrastructure lasts.** What is built for surveillance stays available to future governments and to attackers.

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Research and course

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Thank you

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